CAPTOPRIL

Teratogenic
Empty stomach

CHEMICAL NAME (2S)-1-[(2S)-2-methyl-3-sulfanylpropanoyl]pyrrolidine-2-carboxylic acid
CLASS Angiotensin-converting enzyme (ACE) inhibitor (sulfhydryl-containing)
TRADE NAMES Capoten · Captopril (generic) · Capozide (with HCTZ)

THERAPEUTIC USES / INDICATIONS

- Hypertension, including hypertensive urgency (sublingual/oral, older practice).
- Heart failure.
- Post-myocardial infarction with LV dysfunction.
- Diabetic nephropathy.

ADMINISTRATION

- **PO** 1 h before meals (food ↓ absorption by ~30%).
- Give first dose at bedtime or with patient lying down (first-dose hypotension).
- Short-acting: 2–3 times daily.

DOSAGES

- **Hypertension:** start 12.5–25 mg twice daily; usual 25–50 mg 2–3 times daily; max 150 mg/day.
- **Heart failure:** 6.25–12.5 mg 2–3 times daily, titrated to 50 mg three times daily.
- **Children:** 0.1–0.3 mg/kg 2–3 times daily (specialist).
- Reduce in renal impairment.

PHARMACOKINETICS

- **Onset:** 15–30 min. **Peak:** 1 h. **Duration:** 6–12 h.
- Partly metabolised in liver.
- Excreted in urine. **t½:** ~2 h.

PHARMACODYNAMICS

- Inhibits **ACE** → ↓ angiotensin II → vasodilation, ↓ aldosterone (less Na^+/water retention, K^+ retained).
- ↓ bradykinin breakdown → cough, angioedema.
- Reduces afterload and preload; slows cardiac remodelling.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry persistent cough (bradykinin).
- Taste disturbance (metallic), rash.
- Dizziness, first-dose hypotension.

ADVERSE EFFECTS (SERIOUS)

- Angioedema (face, tongue, airway); can be fatal.
- Hyperkalaemia.
- Acute kidney injury (especially bilateral renal artery stenosis, dehydration, with NSAIDs).
- Neutropenia/agranulocytosis, proteinuria (high doses, collagen disease).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (2nd/3rd trimester fetal toxicity — stop when pregnancy detected).
- History of angioedema.
- Bilateral renal artery stenosis.
- Use with aliskiren in diabetes/renal impairment; with sacubitril (36 h gap).

KEY INTERACTIONS

- **K^+-sparing diuretics, K^+ supplements, trimethoprim:** hyperkalaemia.
- **NSAIDs:** ↓ effect, ↑ AKI.
- **Lithium:** ↑ lithium levels.
- **Diuretics:** first-dose hypotension.

NURSING MANAGEMENT

- Check BP (lying and standing) before each dose.
- Monitor U&E (K^+, creatinine) at baseline and 1–2 weeks after start/dose change.
- Watch for lip/tongue swelling — stop and call for help.
- Hold if SBP <90 or as per prescriber.
- Fall precautions after first dose.

PATIENT EDUCATION

- Take 1 hour before meals.
- Stand up slowly.
- Report swelling of face/lips, persistent cough or dizziness.
- Avoid potassium salt substitutes; use effective contraception.

KEY POINTS

- Short-acting ACE inhibitor; empty stomach.
- Contraindicated in pregnancy.
- Cough, angioedema, hyperkalaemia.
- Check U&E after starting.